Experiments with microwaves

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Abstract

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1 Introduction

The spectrum of electromagnetic radiation covers a wide range of wavelengths, ranging from high-energy γ-rays to low-energy radio waves, all of which play an important role in today's world. Examples include X-rays used in medical diagnostics, the radio waves employed by mobile phones to transmit information, and microwaves used to heat your lunch.
In this experiment, we take a closer look at the latter type of radiation: microwaves. Their wavelengths range from one meter to one millimeter, giving them a wide variety of applications. Furthermore, microwaves play a significant role in our understanding of the universe through the study of the cosmic microwave background, which is believed to be a remnant of the Big Bang, the event that marked the beginning of the universe. Microwaves exhibit many interesting properties. In the following, several of these properties will be examined and discussed, namely the angular distribution and polarization properties of microwaves, their wavelengths, and the formation of evanescent waves. Using these phenomena, we will also determine the lattice constant of a macroscopic cube.

2 Angular distribution of microwaves

To examine the angular distribution of microwaves, the transmitter was placed on a goniometertable, such that the horn is located approximately at the pivot point. The angular distribution was measured to both sides $\pm 90°$ in equidistand steps. The measurements are done with a photo diode that is connected to an Agilent 34402A. The measured intensity was normalized to 1 for $\theta = 0°$ and is shown in fig. **??**. The density of measured points around $\theta = 0°$ is quite low, which is not ideal. One should see the symmetry around the 0-point, but with our

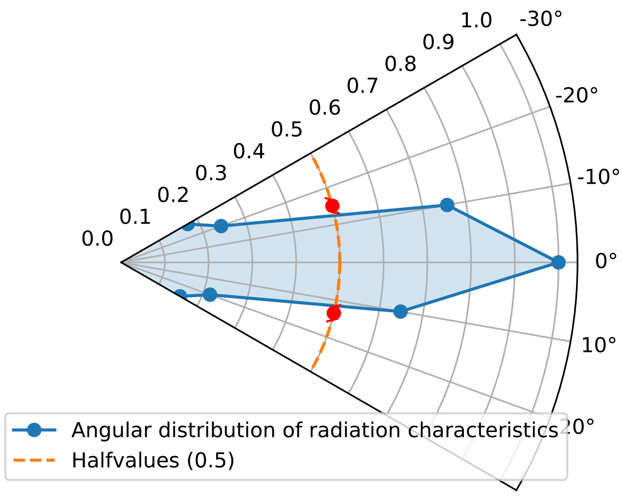

Figure 1: Measured voltage within the relevant angular range normalized to 1 in dependence of the angle θ. The left and right full width at half maximum are marked.

measurements, one can only guess it.
Nevertheless was the angle stretch of the radiation quantified by calculating the halfangle θ_H where the full width at half maximum lies.

$$\theta_H = (28.4 \pm 2.5)° \tag{1}$$

As expected due to the transmitter being a horn antenna, this confirms that the microwave radiation is pointed towards the 0° direction.

References

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